

Quantum-Classical Hybrid Band Selection for Hyperspectral Image Classification

QUBO-formulated band selection solved via simulated annealing, QAOA statevector simulation, and real 30-qubit QAOA on IBM Quantum hardware (ibm_fez) — benchmarked across 8 ML classifiers on the Salinas hyperspectral dataset.

94.85%	0.9426	93%	30 qubits	54,129
OVERALL ACCURACY	KAPPA COEFFICIENT	DIMENSIONALITY REDUCTION	REAL IBM QPU (IBM_FEZ)	PIXELS CLASSIFIED

Approach

Selecting an informative, non-redundant subset of spectral bands from 224 candidates is a combinatorial optimization problem ($C(224,20) \approx 3.5 \times 10^{12}$ combinations) — infeasible to brute-force classically. This project formulates it as a **QUBO**: reward Fisher-discriminant informativeness per band, penalize pairwise correlation, and constrain to exactly k bands selected.

The same QUBO is solved five ways for direct comparison:

- Simulated annealing** — classical baseline, 50 restarts $\times$ 5,000 steps
- QAOA, Qiskit Statevector** — 16-qubit and 20-qubit simulation
- QAOA on real IBM Quantum hardware** — 30 qubits on ibm_fez (156-qubit Heron r2), 15 and 20 selected bands, the latter with 3 $\times$ 3 spatial-spectral feature augmentation

Selected bands then feed 8 classical ML classifiers (SVM, Random Forest, Extra Trees, HistGradientBoosting, KNN, Logistic Regression, MLP, XGBoost) tuned via 5-fold stratified CV on all 54,129 labeled pixels.

Configuration comparison

CONFIG	SOLVER	BANDS	BEST OA
1	Simulated annealing	15	91.06%
2	Statevector, 16 qubits	15	89.19%
3	Statevector, 20 qubits	15	89.19%
4	IBM ibm_fez, real QPU	15	91.46%
5	IBM ibm_fez, real QPU + spatial	20	94.85%

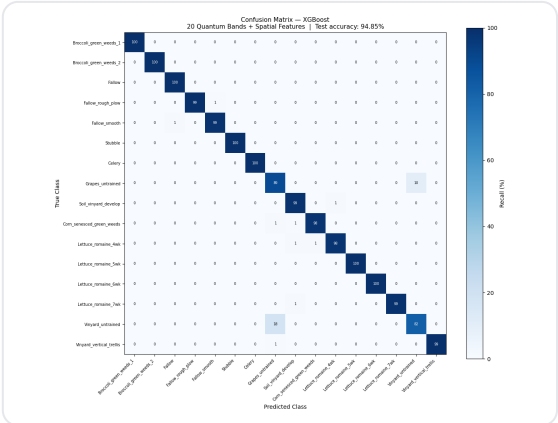
Key findings

Band diversity > qubit count
16q and 20q statevector runs — identical algorithm, restricted candidate pool — converged to the same bands and the same 89.19% accuracy. Qubits alone don't help.

Spatial features: biggest single gain
3 $\times$ 3 neighborhood features on the same 20 IBM-selected bands lifted OA by +3.39pp (91.46% $\rightarrow$ 94.85%) — the largest jump of any change tested.

Real QPU $\approx$ classical optimum, faster
IBM hardware matched simulated annealing's QUBO cost within 0.12%, despite gate noise, in 0.3 min vs. 47 min for 20-qubit statevector simulation.

Confusion matrix — best config



XGBoost, 20 quantum-selected bands + spatial features, 90/10 split, test set n=5,413.

Qiskit 2.0

qiskit-ibm-runtime

IBM ibm_fez · Heron r2

scikit-learn

XGBoost

QAOA

Simulated Annealing